

Lab 01: Introduction to Cisco Packet Tracer & Exploring a Basic Network Topology

Duration: 1 hour 30 minutes | In-Lab Assessment

Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Identify common Packet Tracer device types and explain their intended roles (end devices, network devices, IoT).
- Differentiate common physical and logical ports and modules on devices.
- Choose appropriate cable types for connections and attach cables to correct ports.
- Navigate Packet Tracer workspace: Logical vs Physical view, device box, connection toolbar, and device configuration panels.
- Use Real-Time and Simulation modes to observe basic frame traversal and PDU lifecycles.
- Open and use the built-in device utilities: Desktop apps, CLI/Console, and Programming editor (where available).
- Identify network “edge” and “core” devices.
- Build a LAN topology in Packet Tracer.
- Configure IP addressing and default gateways.
- Generate ICMP (ping) traffic and capture frames.

Lab Description

This lab introduces Cisco Packet Tracer’s interface and device catalogue. You will place devices, inspect their physical ports and built-in utilities, connect devices with appropriate cables, and run simple simulations to observe how Packet Tracer models traffic and device behavior — all without changing IP or advanced routing settings. The focus is on understanding device types, components, ports and software features inside Packet Tracer.

After introduction to the cisco packet tracer software, this lab introduces to the students the basics of internet using a basic network topology, by showing different components.

Part 1: Introduction to Cisco Packet Tracer

Part A: Getting started with Packet Tracer UI

1. Start Cisco Packet Tracer and create a new file.
2. Locate and explore the main UI areas:
 - Device-Type Selection Box (bottom left): categories such as **End Devices**, **Switches**, **Routers**, **Wireless**, **Security**, **Connections**, **Generic**.

- Workspace panes: **Logical** and **Physical** tabs.
 - Device/Connection toolbar (top): **Inspect**, **Delete**, **Move**, **Place Note**.
 - Simulation controls (bottom right): **Real-Time / Simulation** toggle, **Play**, **Capture/-Forward**.
3. Save your file as `Lab1_PT_Intro.pkt`.

Part B: Device catalogue

1. From the **End Devices** category, drag the following and inspect each:
 - **PC-PT**: desktop PC — view **Physical** tab to see NIC ports, power LED, USB/serial approximations; open **Desktop** to see apps (PC GUI, Command Prompt, Web Browser, Programming).
 - **Laptop-PT**: laptop form factor — note built-in wireless NICs in some models and the **Physical** chassis.
 - **Server-PT**: server with virtual services — inspect the **Services** tab (HTTP, FTP, DNS, etc.) and **Config** pages (hostname, passwords).
 - **IP Phone / Generic IoT devices**: briefly inspect for IoT labs.
2. From the **Switches & Router** category:
 - **Switch-PT** (e.g. 2960): view physical ports (FastEthernet/Gigabit)
 - **Router-PT** (e.g. 2911): view physical ports (FastEthernet/Gigabit)
3. From the **Connections** palette, view cable types

Part C: Ports, modules and cable types

1. Click each placed device and open **Physical** (or **Config**) view to inspect available ports:
 - NIC ports labelled **FastEthernet0**, **GigabitEthernet0**, wireless NICs, **USB/Serial** console ports.
2. Cable types and when to use them:
 - **Copper Straight-through**: PC <-> Switch, Server <-> Switch, Access Point <-> Switch. (When the devices are of different types)
 - **Copper Crossover**: PC <-> PC (legacy) or Switch <-> Switch (older devices); Packet Tracer will often auto-select the correct cable. (When the devices are of same types)
 - **Console / RS-232 / Serial**: used for CLI/console access to routers/switches (learn how to connect console cable to device console port and to PC's **Terminal** app).

Part D: Device utilities and software features

1. For each end device (PC/Server), open the **Desktop** tab and explore built-in apps:
 - **Command Prompt** — use it to practice simple Packet Tracer commands that do not require IP (e.g., open the terminal to connect via console).

- **PC GUI / Web Browser / Email / Files** — view how Packet Tracer simulates apps on end devices.
 - **Programming** (on supported devices) — open the editor to inspect example Python templates (do not need to run a networking script in this lab).
 - **Terminal / CLI** — connect console cable from PC to device console and practice opening the console window to see the device prompt.
2. For each network device (Switch/Router), open the CLI tab and explore the basic interface:
- **Command-Line Interface (CLI)** — observe how devices boot up, watch the initial startup messages, and identify the default prompt (**Switch>** or **Router>**).
 - **User EXEC Mode** — practice entering basic commands such as **show ?**, **show version**, **show ip interface brief**.
 - **Privileged EXEC Mode** — enter privileged mode using **enable** and explore common diagnostic commands.
 - **Global Configuration Mode** — navigate using **configure terminal** and locate key configuration contexts such as interface configuration (**interface FastEthernet0/1**).
 - **Interface Exploration** — identify physical and logical interfaces, verify their default status, and understand how Packet Tracer models ports (e.g., shutdown state on routers).
 - **Basic Navigation** — practice moving between modes: User EXEC → Privileged → Global Config → Interface Config, then exit back to CLI.
3. Simulation vs Real-Time:
- Switch to **Simulation Mode** and inspect a PDU as you create it (use the **Add Simple PDU** tool to simulate a generic frame between two devices). Observe PDU lifecycle, event list, and per-hop info.
 - Use Event List Filters to focus on specific protocol families (e.g., ARP, TCP) — you will not need to set IPs. However, you can still observe theoretical PDU processing in the simulated environment.
 - Use **Inspect** on a captured PDU to see fields and how Packet Tracer models headers.

Part 2: Exploring a Basic Network Topology

Part E: Packet Tracer Basics

1. Components

- **Devices Palette:** End Devices → PC-PT.
- **Network Devices:** Switches → 2960; Routers → 2901.
- **Connections:** Click the lightning-bolt icon for cables (copper straight-through).

2. Save Your File: File → Save As → Lab1_Topology.pkt.

Part F: Building the Topology

1. Place Devices:

- PC1: drag PC-PT (left).
- Switch: drag Switch-2960 (center).
- Router: drag Router-2901 (right of switch).
- PC2: drag PC-PT (right).

2. Connect Devices:

- PC1 (FastEthernet0) → Switch (FastEthernet0/1) using Copper Straight-Through.
- Switch (FastEthernet0/2) → Router (GigabitEthernet0/0).
- Router (GigabitEthernet0/1) → PC2 (FastEthernet0) using Copper Straight-Through.

Part G: IP Configuration

The IP is configured on the interface.

1. PC1:

- Open PC1 → Desktop → IP Configuration:
 - IP: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1

2. Router:

In Packet Tracer, click Router → **CLI** tab, then execute:

```
1 enable ! Enter privileged EXEC mode
2 configure terminal ! Enter global configuration mode
3 interface GigabitEthernet0/0 ! Select interface G0/0
4 ip address 192.168.1.1 255.255.255.0 ! Assign IP
5 no shutdown ! Bring up interface
6 exit ! Exit to global mode
7 interface GigabitEthernet0/1 ! Select interface G0/1
8 ip address 10.0.0.1 255.255.255.0 ! Assign IP
9 no shutdown
10 exit
11 end ! Return to privileged EXEC
```

3. PC2:

- Open PC2 → Desktop → IP Configuration:
 - IP: 10.0.0.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 10.0.0.1

Part H: Router-to-Router Connections (Ethernet or Serial)

Routers can be connected to each other using **Ethernet interfaces** or **Serial interfaces**. Serial interfaces are *not required* unless we are using traditional WAN technologies.

Option 1: Router-to-Router Using Ethernet (Recommended)

- No additional modules are needed.
- Use a Copper Straight-Through cable.
- Connect GigabitEthernet0/1 of Router1 to GigabitEthernet0/1 of Router2.

Example Configuration:

```
1  ! Router1
2  enable
3  configure terminal
4    interface GigabitEthernet0/1
5      ip address 172.16.0.1 255.255.255.252
6      no shutdown
7  end
8
9  ! Router2
10 enable
11 configure terminal
12   interface GigabitEthernet0/1
13     ip address 172.16.0.2 255.255.255.252
14     no shutdown
15 end
```

This is simpler and represents modern networks. Remember, both of the routers need to be in the same network

Option 2: Router-to-Router Using Serial Ports

To add serial interfaces:

- Click Router → **Physical** tab.
- Power off router.
- Insert HWIC-2T serial module.
- Power on router.
- Connect Serial10/0/0 <-> Serial10/0/0 using DCE/DTE cable.

Example Configuration:

```
1  ! Router1
2  interface Serial10/0/0
3    ip address 172.16.0.1 255.255.255.252
4    no shutdown
5
6  ! Router2
7  interface Serial10/0/0
```

```

8 ip address 172.16.0.2 255.255.255.252
9 no shutdown

```

Part I: Adding More Ports to a Router Using Modules

Routers such as the 2901 have limited built-in interfaces. Packet Tracer allows you to add additional ports (Ethernet, Serial, etc.) using hardware modules.

Steps to Add a Module:

1. Click the router → go to the **Physical** tab.
2. Power off the router using the power switch (top-left).
3. Drag a module into an empty expansion slot:
 - HWIC-2T — adds 2 serial interfaces.
 - HWIC-4ESW — adds 4 switch-like Ethernet ports.
 - HWIC-1GE-SFP — adds 1 GigabitEthernet port.
4. Power the router back on.
5. Go to the **CLI** tab and verify new interfaces using:

```

1 show ip interface brief

```

Example: Configuring a New Ethernet Port (G2/0)

```

1 enable
2 configure terminal
3 interface GigabitEthernet2/0
4 ip address 192.168.50.1 255.255.255.0
5 no shutdown
6 exit
7 end

```

After adding modules, you can connect more PCs, routers, or switches to these interfaces.

Part J: Establishing Routing (RIP)

Configure RIP on each router via the CLI:

```

1 enable
2 configure terminal
3 router rip ! Enable RIP process
4 version 2 ! Use classless RIP v2
5 network 192.168.1.0 ! Advertise LAN on G0/0
6 network 10.0.0.0 ! Advertise LAN on G0/1
7 network 172.16.0.0 ! Advertise serial link subnet
8 exit
9 end

```

Configure using GUI:

- Click the router and then click on config

- Then select RIP from the left side section
- There you can see an field to add network address and a button add
- Add all the addresses a router is connected to (including the router to router network).
- Do this in both the routers

Part K: Simulation & Packet Capture

- Switch Packet Tracer to **Simulation Mode**.
- On PC1 (Desktop → Command Prompt), run:
ping 10.0.0.10 ! Send ICMP echo to PC2
- Observe the ICMP frames traversing the network

Part L: Saving Configurations

To save your changes:

```
1 enable ! Ensure privileged EXEC mode
2 copy running-config startup-config ! Save running config to NVRAM
```